4 int main ( )

C++ program is a collection of functions. Every C++ program must have a main function. The main( ) function is the starting point where all C++ programs begin their execution. Therefore, the executable statements should be inside the main( ) function.

```
5  {
6      cout << "Welcome to Programming in C++";
7      return 0;
8  }
```

The statements between the curly braces (Line number 5 to 8) are executable statements. This is actually called as a block of code. In line 6, cout simply sends the string constant "Welcome to Programming in C++" to the screen. As we discussed already, every executable statement must terminate with a semicolon. In line 7, return is a keyword which is used to return the value what you specified to a function. In this case, it will return 0 to main( ) function.

9.6 Execution of C++ program:

For creating and executing a C++ program, one must follow four important steps.

(1) Creating Source code

Creating includes typing and editing the valid C++ code as per the rules followed by the C++ Compiler.

(2) Saving source code with extension .cpp

After typing, the source code should be saved with the extension .cpp

(3) Compilation

This is an important step in constructing a program. In compilation, compiler links the library files with the source code and verifies each and every line of code. If any mistake or error is found, it will throw error message. If there are no errors, it translates the source code into machine readable object file with an extension .obj

(4) Execution

This is the final step of a C++ Program. In this stage, the object file becomes an executable file with extension .exe. Once the program becomes an executable file, the program has an independent existence. This means, you can run your application without the help of any compiler or IDE.

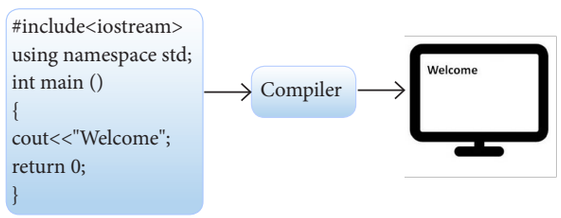


Figure 9.8 Execution

9.7 C++ Development Environment

There are lot of IDE programs available for C++. IDE makes it easy to create, compile and execute a C++ program. Most of the IDEs are open source applications (ie.) they are available free of cost.

9.7.1 Familiar C++ Compilers with IDE

Table 9.4 Open Source Compilers

Compiler	Availability
Dev C++	Open source
Geany	Open source
Code::blocks	Open source
Code Lite	Open source

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